

Team-discussion prompts: Linguistic Analysis and Information Extraction

Topic A: Named entity recognition

Task 1: What are different types of NER and what are their properties?
(for example, flat or nested)

Task 2: Read a top-tier AAAI conference paper published in 2022 on how NER can be used for an actual application. Look through the Github code and explain your brief understanding of the article.

Resources:

- <https://github.com/ljynlp/W2NER?tab=readme-ov-file>
- <https://arxiv.org/pdf/2112.10070.pdf>

Topic B: Parts-of-speech tagging

Task 1: How is it possible to do keyphrase extraction using POS tagging? Try out chunking of noun phrases and verb phrases using regular expressions

Resources:

- <https://towardsdatascience.com/nlp-and-pos-based-chunking-to-generate-amazon-style-key-phrases-from-reviews-239084543225>
- <https://suneelpatel18.medium.com/nlp-pos-part-of-speech-tagging-chunking-f72178cc7385>

Task 2: Show a demo code of Task 1 to extract different type of phrases using any sample dataset

Topic C: Dependency parsing

- Task 1:** Explain which is the correct dependency parsing tree for the following sentence "She took the lesson to heart"
- Task 2:** Which of the following trees for the sentence "workers dumped sacks of garbage and junk into a bin" are correct? (there are multiple correct answers - you may use Python to generate the trees)

